

Examen Final

1) $I = 1 \text{ kgm}^2$ $R = 0.3 \text{ m}$ $K = 50 \text{ N/m}$ $m = 0.5 \text{ Kg}$
 $d = 0.2 \text{ m}$ $\theta = 37^\circ$

$$K_o + U_o + U_{s_o} + K_{R_o} = K_e + U_f + U_s + K_{R_f}$$

$$mg h_o + \frac{1}{2} K x^2 = \frac{1}{2} m V_f^2 + \frac{1}{2} I \omega^2$$

$$mg \sin \theta \cdot d + \frac{1}{2} K d^2 = \frac{1}{2} m \omega^2 R^2 + \frac{1}{2} I \omega$$

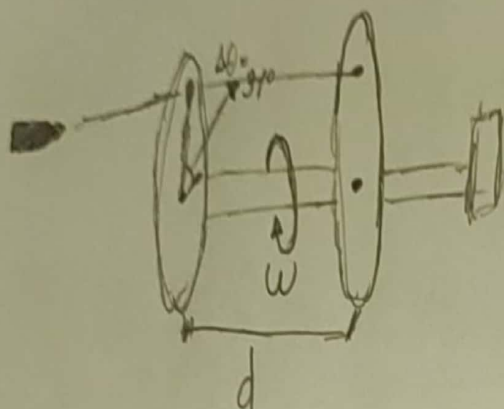
$$2mg \sin \theta d + K d^2 = \omega^2 (m R^2 + I)$$

$$\omega = \sqrt{\frac{2mg \sin \theta d + K d^2}{m R^2 + I}}$$

$$\omega = \sqrt{\frac{2(0.5)(9.8) \sin 37^\circ (0.2) + 50(0.2)^2}{(0.5)(0.3)^2 + 1}}$$

$$\omega = \underline{1.74 \text{ rad/s}}$$

2)



$$d = 80 \text{ cm} = 0.8 \text{ m}$$

$$\omega = 900 \frac{\text{rev}}{\text{min}}$$

$$\Delta\theta = 31^\circ$$

$$\omega = \frac{\Delta\theta}{t}$$

$$v = \frac{d}{t}$$

$$t = \frac{\Delta\theta}{\omega}$$

$$v = \frac{\omega d}{\Delta\theta}$$

$$v = \frac{900 \frac{\text{rev}}{\text{min}} \cdot \frac{360^\circ}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ sec}} \cdot 0.8 \text{ m}}{31^\circ} = 139.35 \approx 139.4$$

$$4) \quad m = 10g$$

$$m = 0.01 \text{ Kg}$$

$$x = ?$$

$$m_b = 2 \text{ Kg}$$

$$K = 19.6 \text{ N/m}$$

$$w = 300 \text{ m/s}$$

$$V = \frac{0.01 \times 300}{0.01 \times 2} = 1.49 \text{ m/s}$$

$$\frac{1}{2} m_b V^2 = \frac{1}{2} x V^2$$

$$x = \sqrt{\frac{m_b + m}{K} V^2} = \sqrt{\frac{m_b + m}{K}} V$$

$$x = \sqrt{\frac{(2 + 0.01)}{19.6}} \times 1.49 = 0.4771$$

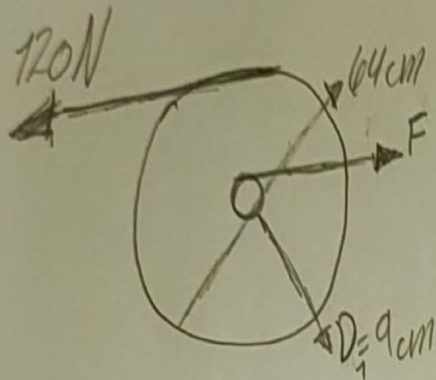
$$x = 0.48 \text{ m}$$

5)

$$D_2 = 64 \text{ cm}$$

$$m = 1.8 \text{ kg}$$

$$F = 120 \text{ N}$$



$$\alpha = 45 \text{ rad/s}^2$$

$$I = mR^2$$

$$\frac{64}{2} = 32 \text{ cm}$$

$$0.32 \text{ m}$$

$$\frac{9}{2} = 4.5 \text{ cm}$$

$$0.045 \text{ m}$$

$$\tau_{\text{net}} = \tau_F - \tau_{Fr}$$

$$I\alpha = FR_1 - 120N(R_2)$$

$$(mR^2)\alpha = F\left(\frac{D_1}{2}\right) - 120N\left(\frac{D_2}{2}\right)$$

$$(1.8 \times 0.32^2)4.5 = F(0.045) - 120N(0.32)$$

$$0.829 \text{ Kg m}^2/\text{s}^2 = F(0.045) - 38.4 \text{ N/m}$$

$$(0.829 + 38.4) \text{ N/m} = F(0.045)$$

$$39.229 \text{ N/m} = F$$

$$0.045$$

$$F = 871.76 \text{ N}$$